HELM

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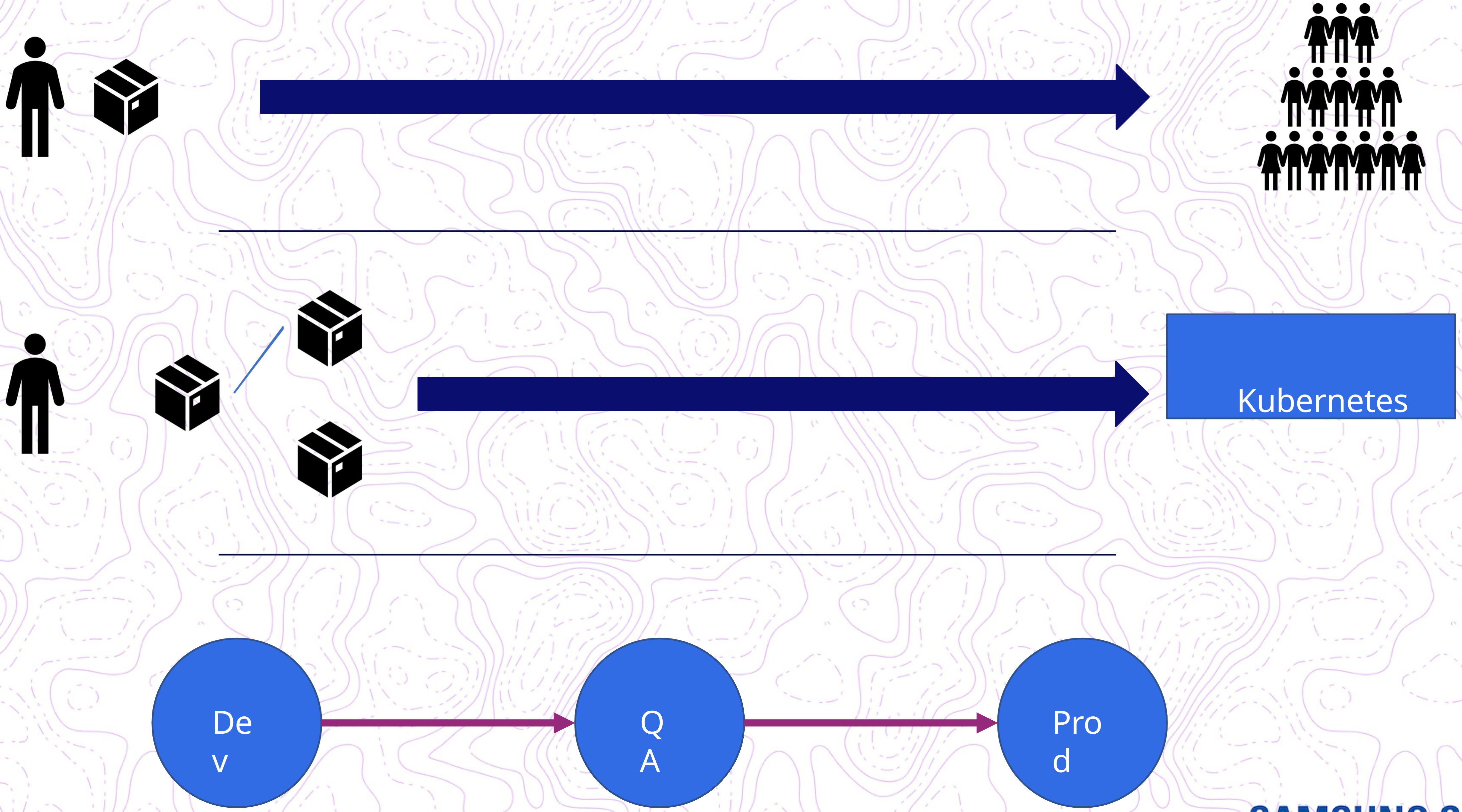
Helm Is The Package Manager For Kubernetes

Helm 은 Kubernetes 의 패키지 관리자입니다

≈

Apt



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You Can Download And Run Helm For These Systems
다양한 OS 별 실행파일을 지원

macOS

Windows

3

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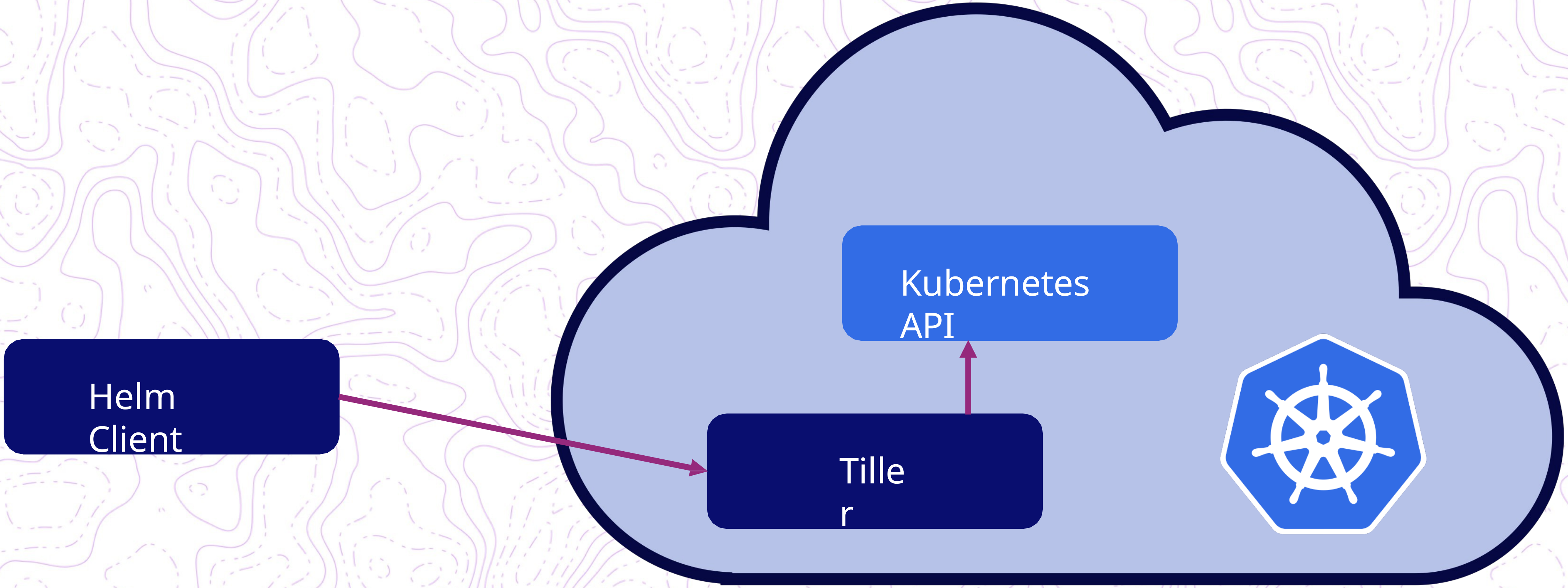
A Similar Command Line Experience
예전과 유사한 커맨드 라인

```
$ helm repo add stable https://kubernetes-charts.storage.googleapis.com/
$ helm search repo mariadb
NAME                 CHART VERSION      APP VERSION        DESCRIPTION
stable/mariadb       7.1.0              10.3.20            Fast, reliable, scalable, and easy to use open-...
stable/phpmyadmin    4.2.4              4.9.2              phpMyAdmin is an mysql administration frontend
$ helm install mymaria stable/mariadb
$ helm list
NAME      NAMESPACE REVISION   UPDATED                                  STATUS     CHART           APP
                                                                                   VERSION
mymaria   default      1       2019-11-27 14:10:55.216169 -0500 EST     deployed   mariadb-7.1.0   10.3.20
$ helm uninstall mymaria
```

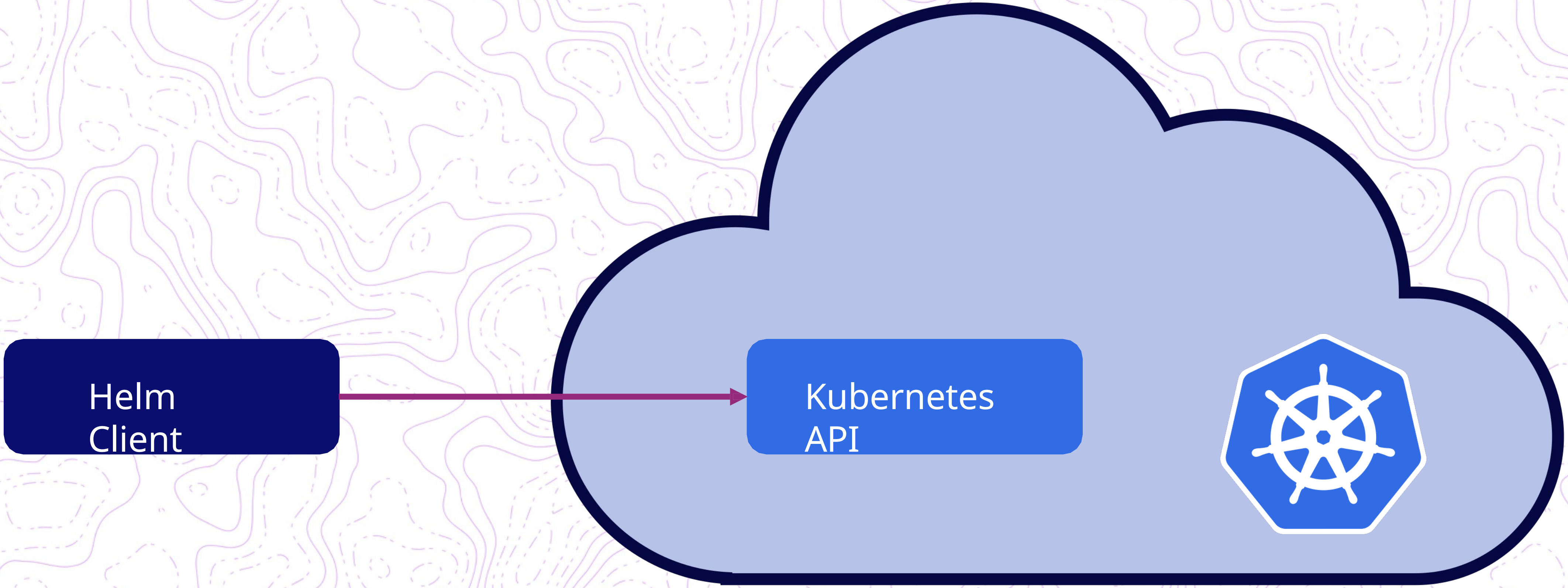
Namespaces!
네임 스페이스!

```
$ helm list
NAME       NAMESPACE REVISION  UPDATED                                  STATUS    CHART              APP VERSION
mymaria    default    1        2019-11-27 14:10:55.216169 -0500 EST    deployed  mariadb-7.1.0      10.3.20
$ helm ls --all-namespaces
NAME       NAMESPACE REVISION  UPDATED                                  STATUS    CHART              APP VERSION
mymaria    default    1        2019-12-01 18:05:53.568566 -0500 EST    deployed  mariadb-7.1.0      10.3.20
mymaria2   foo        1        2019-12-01 18:06:12.520784 -0500 EST    deployed  mariadb-7.1.0      10.3.20
$ helm ls —-namespace foo
NAME       NAMESPACE REVISION  UPDATED                                  STATUS    CHART              APP VERSION
mymaria2   foo        1        2019-12-01 18:06:12.520784 -0500 EST    deployed  mariadb-7.1.0      10.3.20
$ helm install –n foo myNewMaria stable/mariadb
```

Helm v2 Architecture
Helm v2 아키텍처



Helm v3 Architecture
Helm v3 아키텍처



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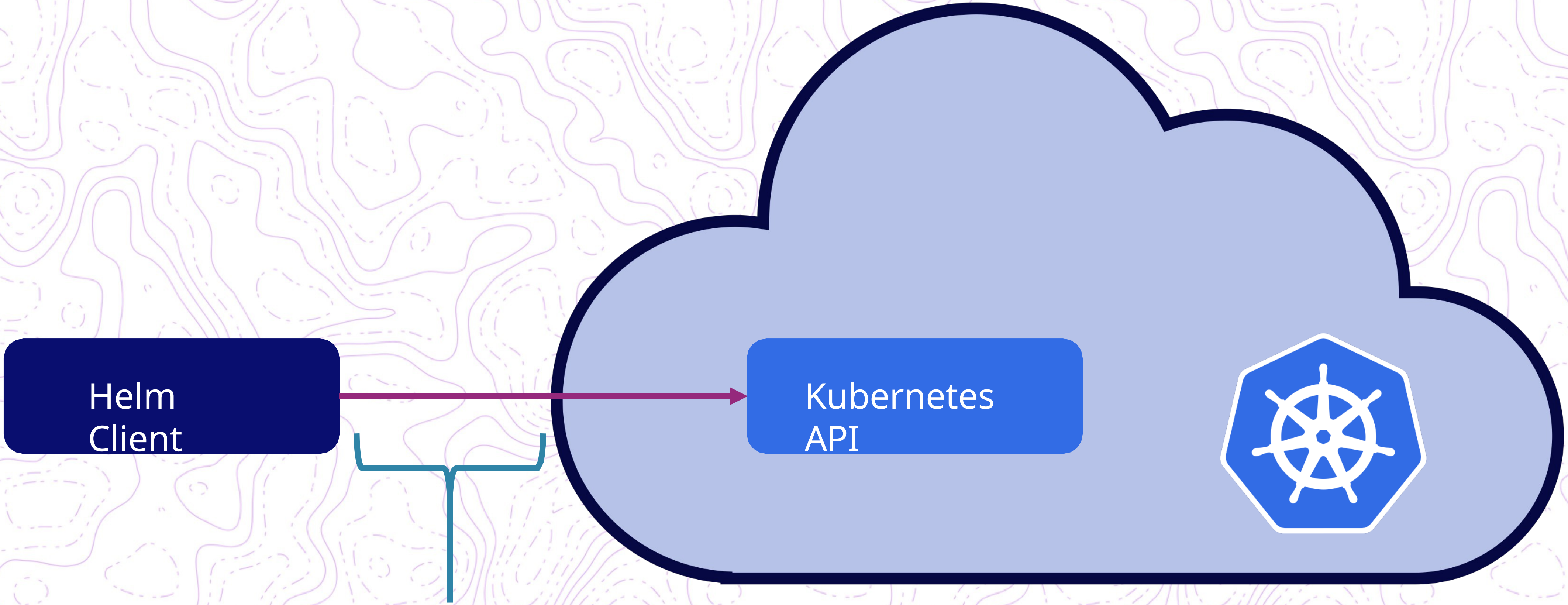
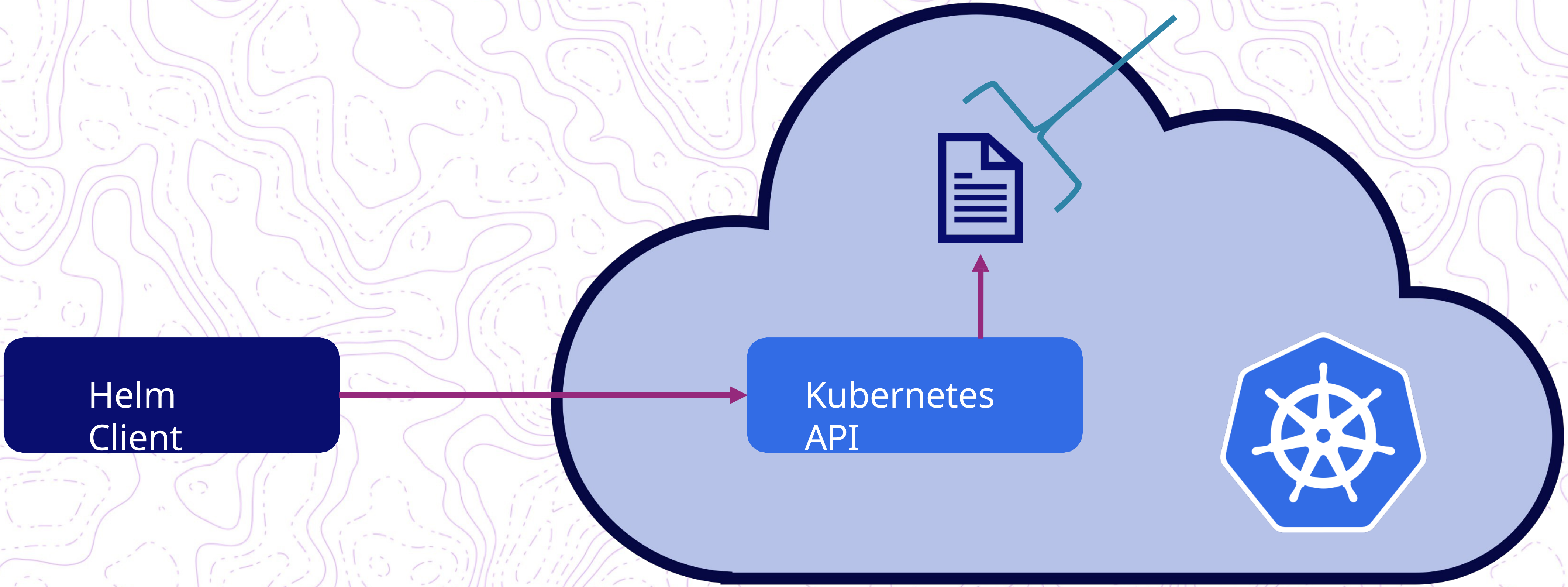
Helm v3 Architecture
Helm v3 아키텍처



Uses normal user credentials
일반 사용자 인증 사용

Helm v3 Architecture
Helm v3 아키텍처

Release information stored as *Secret* in the namespace with the app by default 앱이 설치된 네임스페이스에 릴리즈 정보가 *Secret* 으로 저장 (기본 설정)



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helm.sh/docs/faq/#xdg-base-directory-support

You are viewing info for Helm 3 - check the **version FAQs** or **return to Helm 2** for prior versions.

H ELM Documentation

G et Helm B1og D ocs Charts v3.o. o st able

Docs Home
Introduction
How-To

Topics

Best Practices
Developing Templates

Community
Frequently Asked Questions

Glossary

G XD Base Directory Support

The XDG Base Directory Specification is a portable standard defining where configuration, data, and cached files should be stored on the filesystem.

In Helm 2, Helm stored all this information in -/ . helm (affectionately known as helm home), which could be changed by setting the 4Hz£N Honz environment variable, or by using the global flag --home .

In Helm 3, Helm now respects the following environment variables as per the XDG Base Directory Specification:

- $XDG_CAC HE_HOFIE
- $XDG_CONFIG_HOME
- $XDG_DATA_HOME

Helm plugins are still passed $HELM EOuE as an alias to $xDC DA'IA HOI‹z for backwards compatibility with plugins looking to use $nEkM I4OME aS a scratchpad environment.

Several new environment variables are also passed in to the plugin s environment to accommodate this change:

- $HEI M PA'I'II CACHz for th e cache path
- $HEI M_PA'I'H_CONE'IG fOr the config path
- $HELM PATH DATA forthedata path

Helm plugins looking to support Helm 3 should consider using these new environment variables instead.

O Contribute to D ocs

helm.sh/helm/v3/pkg /D

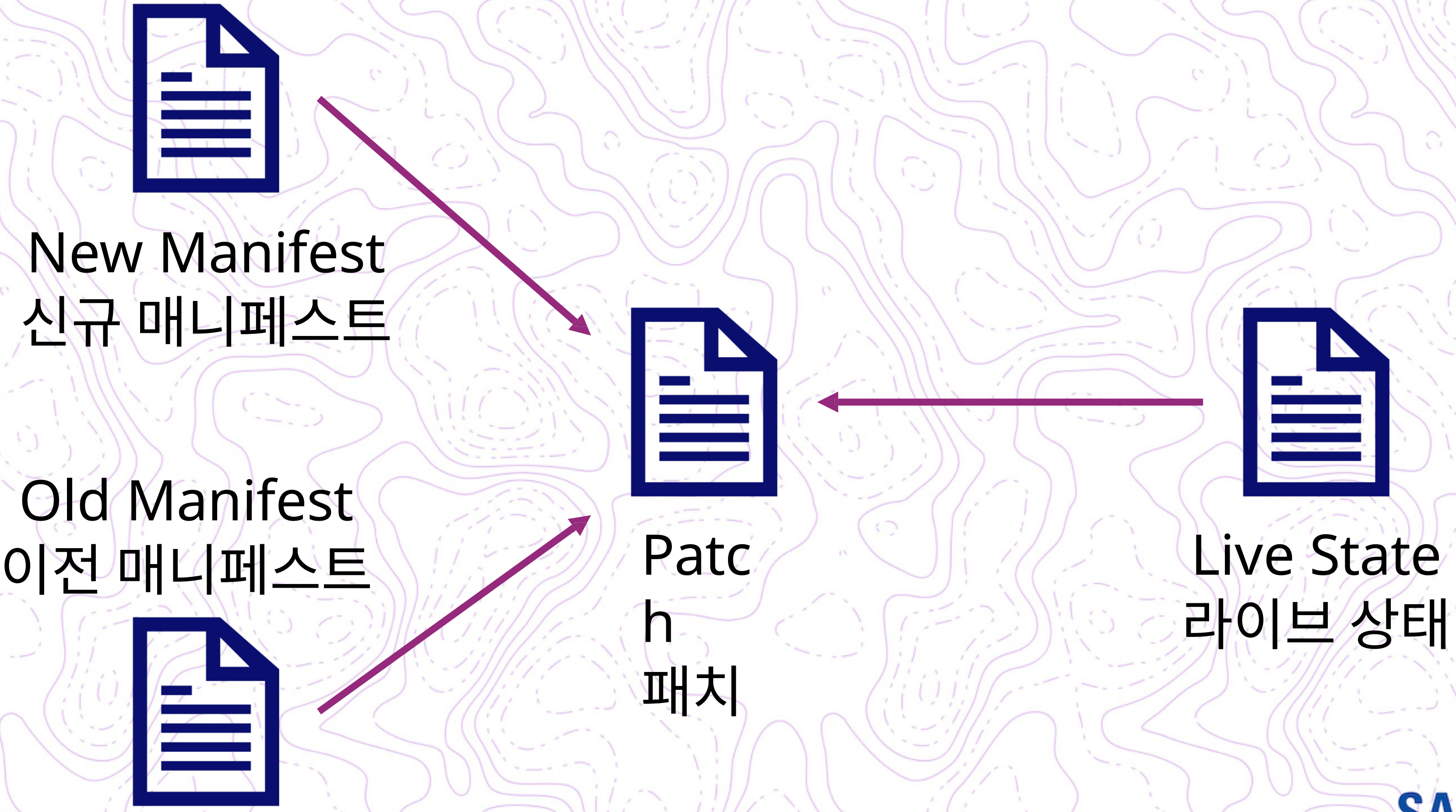
directory helm.sh/he\m/v3/pkg v300

Published: Nov 12, 2019 | License: Apaohe-2.0 | Module: helm.sh/helm/v3

Doc By Overview Subdirectories Versions Imports Imported Licenses

Path	Synopsis
action	Package action contains the logic for each action that Helm can perform.
chart	
chart/loader	
chartutil	Package chartutil contains tools for working with charts.
cli	Package cli describes the operating environment for the Helm CLI.
cli/output	
cli/values	
downloader	Package downloader provides a library for downloading charts.
engine	Package engine implements the Go template engine as a Tiller Engine.
gates	Package gates provides a general tool for working with experimental feature gates.
getter	Package getter provides a generalize tool for fetching data by scheme.
helmpath	Unless required by applicable law or agreed to in writing, software distributed under the License is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
helmpath/xdg	
kube	
kube/fake	Package fake implements various fake KubeClients for use in testing

3-way Strategic Merge Patches
3 방향 전략적 병합 패치



3-way Strategic Merge Patches
3 방향 전략적 병합 패치

```
containers:
- name:  server
  image: nginx:2.0.0
```

```
containers:
- name:       server
  image: nginx:2.0.0
- name:       my-injected-
  sidecar image: my-cool-
  mesh:1.0.0
```

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Additions To Charts
차트내 신규 기능 추가

```
$ tree newChart

newChart
├── Chart.yaml
├── charts
├── templates
│    ├──
NOTES.txt
│    ├──
_helpers.tpl
│    ├──
deployment.yaml
│    ├──
ingress.yaml
│    ├──
service.yaml
│    ├──
serviceaccount.
yaml
│    └── tests
│         └──
test-
connection.yaml
└── values.yaml
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: {{ include "newChart.fullname" . }}
  labels:
    {{- include "newChart.labels" . |
    nindent 4 }}
spec:
  replicas: {{ .Values.replicaCount }}
...
```

apiVersion v2 For Helm 3
Helm 3 용 apiVersion v2

Chart.yaml

```yaml
apiVersion: v2
name: mychart
description: A Helm chart for Kubernetes
```

Dependencies Now In Chart.yaml
현재 : Chart.yaml 의 종속성

Chart.yaml

```yaml
apiVersion: v2
name: mychart
description: A Helm chart for Kubernetes

dependencies:
- name: mariadb
  version: 5.x.x
  repository: https://kubernetes-charts.storage.googleapis.com/
  condition: mariadb.enabled
  tags:
    - database
```

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Library Charts

Chart.yaml

```yaml
apiVersion: v2
name: mychart
description: A Helm chart for Kubernetes
type: library
```

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JSON Schema Validation
JSON 스키마 유효성 검사

values.schema.json

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "properties": {
    "addresses": {
      "description": "List of addresses",
      "items": {
        "properties": {
          "city": {
          "type":
            "string"
        },
...
```

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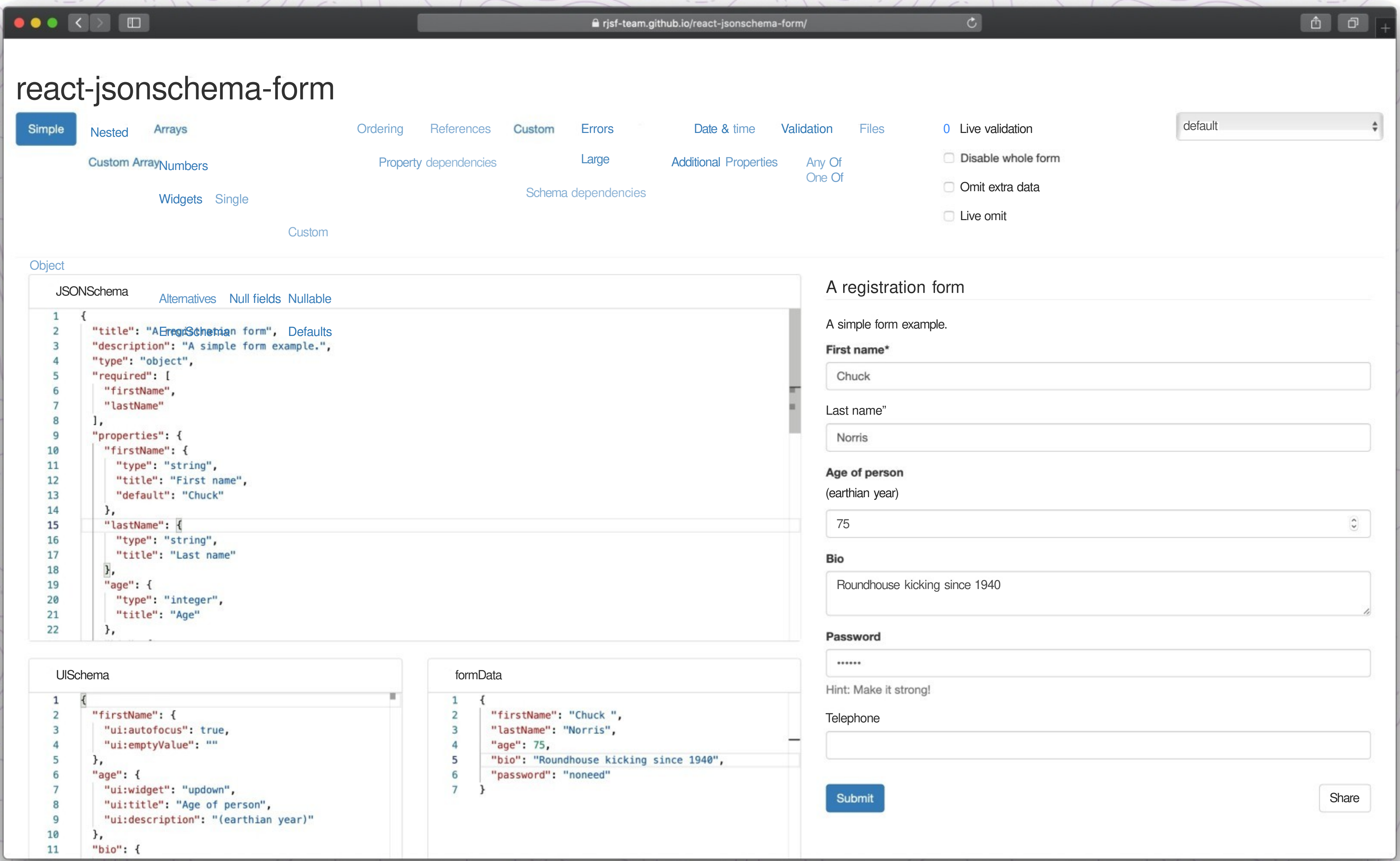

rjsf-team.github.io/react-jsonschema-form/
react-jsonschema-form
Simple
Nested
Arrays
Ordering
References
Custom
Errors
Date & time
Validation
Files
Live validation
default
Custom Array
Numbers
Property dependencies
Large
Additional Properties
Any Of
One Of
Disable whole form
Omit extra data
Live omit
Schema dependencies
Widgets
Single
Custom
Object
JSONSchema
Alternatives
Null fields
Nullable
ErrorSchema
Defaults
{
"title": "A registration form",
"description": "A simple form example.",
"type": "object",
"required": [
"firstName",
"lastName"
],
"properties": {
"firstName": {
"type": "string",
"title": "First name",
"default": "Chuck"
},
"lastName": {
"type": "string",
"title": "Last name"
},
"age": {
"type": "integer",
"title": "Age"
},
UISchema
{
"firstName": {
"ui:autofocus": true,
"ui:emptyValue": ""
},
"age": {
"ui:widget": "updown",
"ui:title": "Age of person",
"ui:description": "(earthian year)"
},
"bio": {
formData
{
"firstName": "Chuck ",
"lastName": "Norris",
"age": 75,
"bio": "Roundhouse kicking since 1940",
"password": "noneed"
}
A registration form
A simple form example.
First name*
Chuck
Last name*
Norris
Age of person
(earthian year)
75
Bio
Roundhouse kicking since 1940
Password
Hint: Make it strong!
Telephone
Submit
Share

Custom Resource Definitions (CRD)

```
$ tree -L 1 ambassador
ambassador
├── CHANGELOG.md
├── Chart.yaml
├── OWNERS
├── README.md
├── ci
├── crds
├── templates
└── values.yaml
```

```
$ tree ambassador/crds
ambassador/crds
├── authservice.yaml
├──
consulresolver.yaml
├── filter.yaml
├── filterpolicy.yaml
├──
kubernetesendpointreso
lver.yaml
├──
kubernetesserviceresol
ver.yaml
├── mapping.yaml
├── module.yaml
├── ratelimit.yaml
├──
ratelimitservice.yaml
├── tcpmapping.yaml
├── tlscontext.yaml
└──
tracingservice.yaml
```

CRDs across Helm v2 and v3
Helm v2 및 v3의 CRD

For Helm v3 Use The *crds* Directory Helm v3 의 경우 crds 디렉토리를 사용하십시오 .

```
$ tree ambassador/crds
ambassador/crds
├── authservice.yaml
├──
consulresolver.yaml
├── filter.yaml
├── filterpolicy.yaml
├──
kubernetesendpointreso
lver.yaml
├──
kubernetesserviceresol
ver.yaml
├── mapping.yaml
├── module.yaml
├── ratelimit.yaml
├──
ratelimitservice.yaml
├── tcpmapping.yaml
```

For Helm v2 Use A Template With *crd-install* Hook Helm v2 의 경우 crd 설치 후크가있는 템플릿 사용

```
apiVersion: apiextensions.k8s.io/v1beta1
kind: CustomResourceDefinition
metadata:
  name: authservices.getambassador.io
  labels:
    app.kubernetes.io/name:
    ambassador
  annotations:
    "helm.sh/hook": crd-install
...
```

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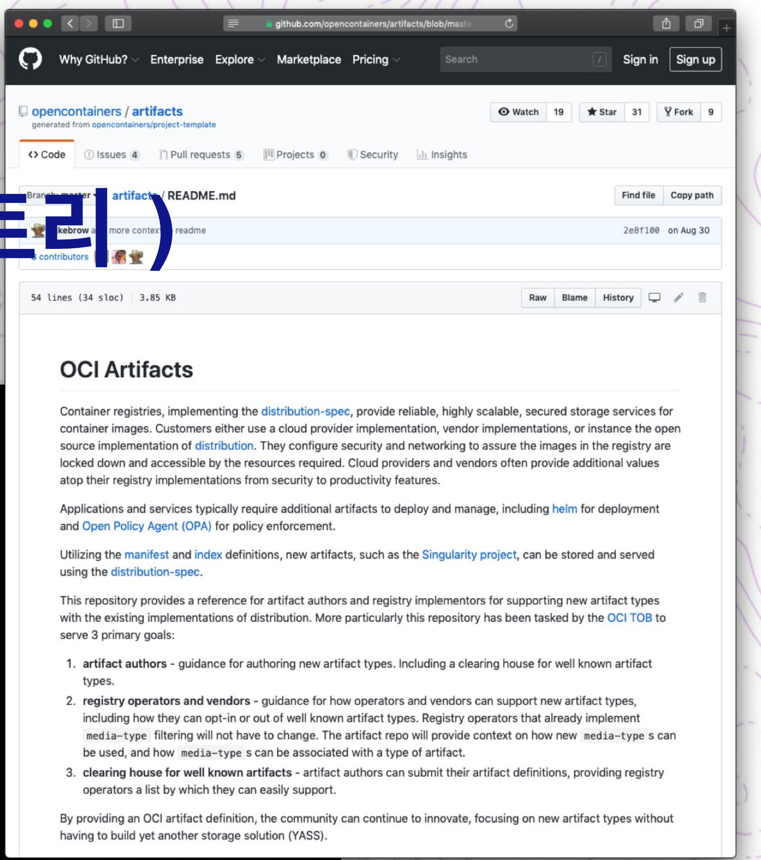
Helm Experiments (베타 기능) OCI Registries (OCI 레지스트리)




```
$ export HELM_EXPERIMENTAL_OCI=1
$ docker run -dp 5000:5000 --restart=always --name registry registry
$ helm chart save mychart localhost:5000/myrepo/mychart:1.2.3
ref:     localhost:5000/myrepo/mychart:1.2.3
digest:  896935a875c8fe8f8b9b81e5862413de316f8da3d6d9a7e0f6f1e90f6204f551
size:    2.7 KiB
name:    mychart
version: 0.1.0
1.2.3: saved
$ helm chart list
REF                                     NAME     VERSION  DIGEST   SIZE     CREATED
localhost:5000/myrepo/mychart:1.2.3     mychart  0.1.0    896935a  2.7 KiB  About a minute
```

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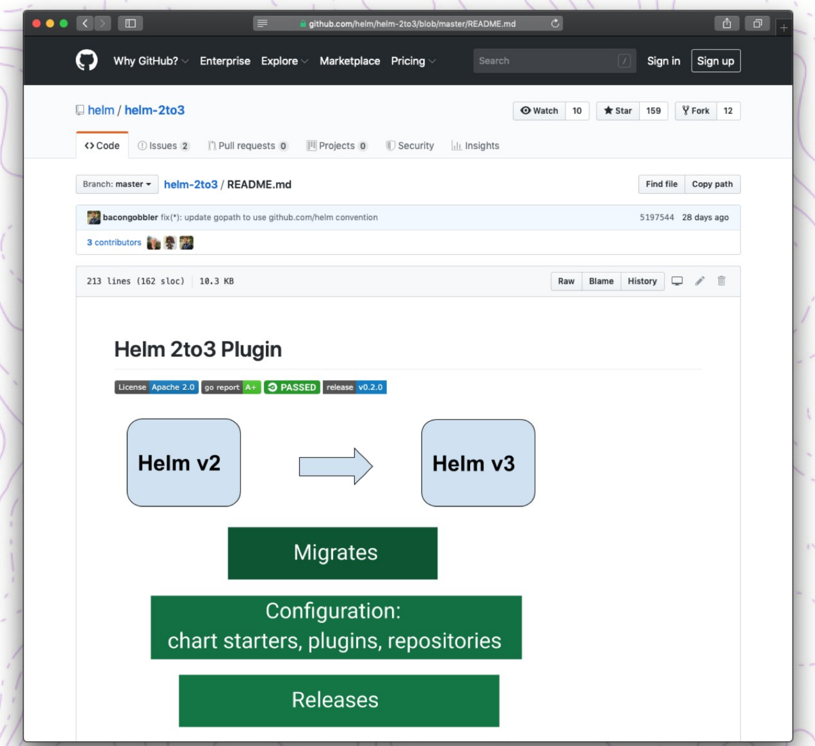
What about upgrading to Helm 3?
Helm 3 으로 업그레이드는 어떻게 할 수 있나요 ?

After all, the storage format changed

```
$ helm plugin install https://github.com/helm/helm-2to3
$ helm 2to3 move config
$ helm 2to3 convert RELEASE
$ helm 2to3 cleanup
```



* Read the docs to learn about flags like --dry-run

--dry-run 과 같은 플래그는 docs 를 확인해주세요

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Learn more: https://helm.sh

Read Docs: https://helm.sh/docs

Source:

https://github.com/helm/helm

Mailing list: cncf-helm@lists.cncf.io

https://lists.cncf.io/g/cncf-helm

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